Solution to Tutorial exercises: Week 4.

Exercise: PV Hosting Capacity and Volt-Watt Control using the Test LV Circuit

E.1: Starting from 0% PV penetration, increase the penetration by 20%, up to 100% PV penetration for **each** of the given days per season (also using the pre-selected PV profile). Test the three Volt-Watt curves (settings) below and check the customer voltages and the resulting curtailment. Based on that, decide which one is the **best curve**.

- **Summer Day: start_day=28**
- **Autumn Day: start_day=54**
- Curve 1:
  - **X_array = "(0.500, 1.090, 1.100, 1.500)"**
  - **Y_array = "(1.0, 1.0, 0.0, 0.0)"**
- Curve 2:
  - **X_array = "(0.500, 1.095, 1.100, 1.500)"**
  - **Y_array = "(1.0, 1.0, 0.0, 0.0)"**
- Curve 3:
  - **X_array = "(0.500, 1.095, 1.104, 1.500)"**
  - **Y_array = "(1.0, 1.0, 0.0, 0.0)"**

Solution:

To answer this question, we first plot the customer voltages. Figures 1 and 2 show the voltages for the summer day and the autumn day, respectively. Each row corresponds to a PV penetration level from 0% to 100% in 20% steps. From these plots, we can see that all three Volt-Watt curves keep the voltages below the 1.1 pu limit. However, it is not easy to tell from the voltage plots alone which curve performs best.

To determine the best curve, we compare the total curtailed energy. For each PV penetration level, we run two power-flow cases: (1) without inverter control and (2) with Volt-Watt control enabled. Because the time step is 30 minutes, the curtailed energy at each step is simply the difference in PV output between the two cases. We then sum these differences over the day to obtain the total curtailed energy. Figures 3 and 4 show the results and report the total curtailed energy in the top-left corner of each subplot. Based on these values, Curve 3 is the best option because it results in the lowest curtailment.

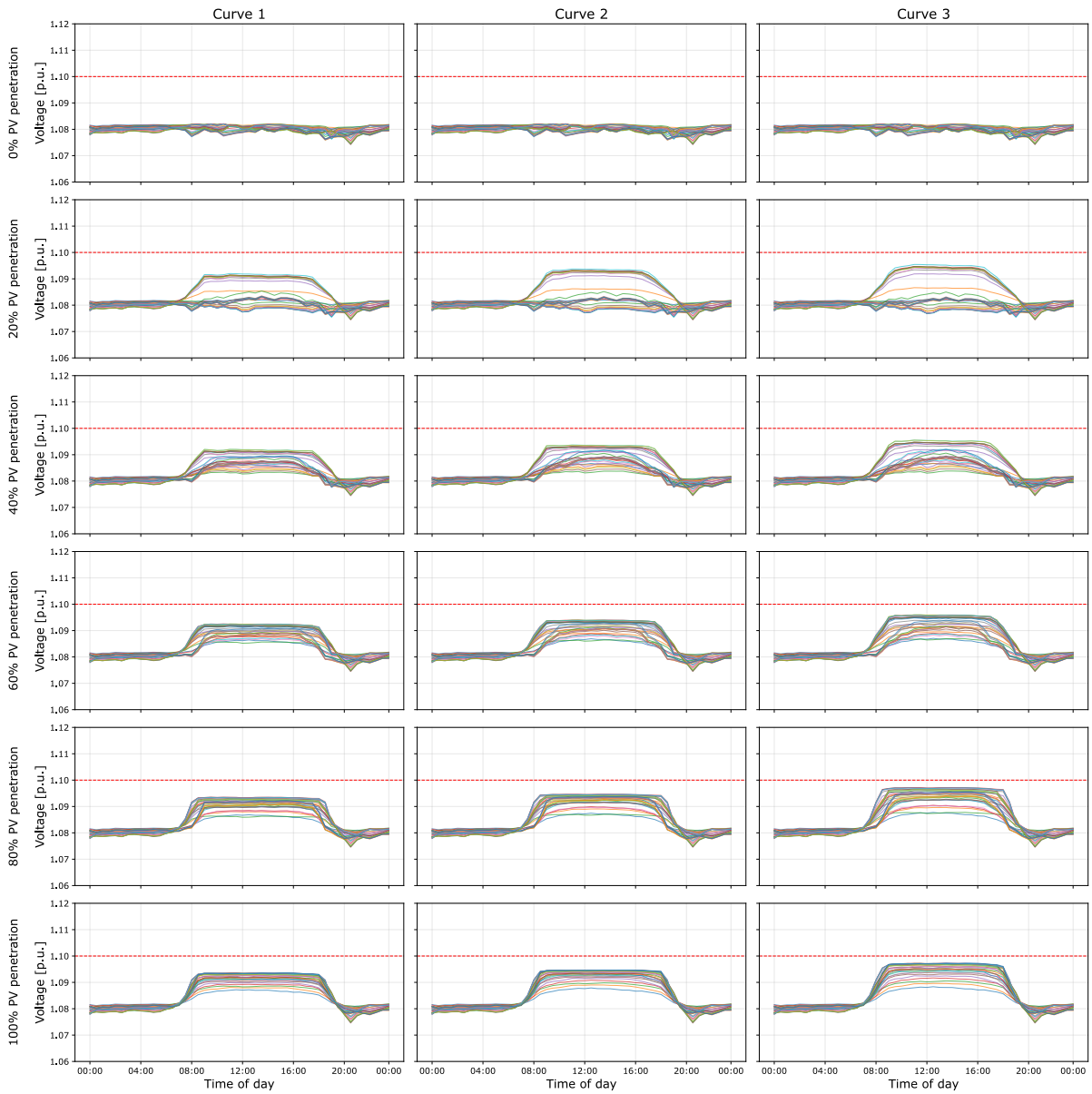

Figure 1: Customer voltage profiles (Summer day, start_day = 28) for PV penetration levels from 0% to 100% and Volt-Watt Curves 1–3.

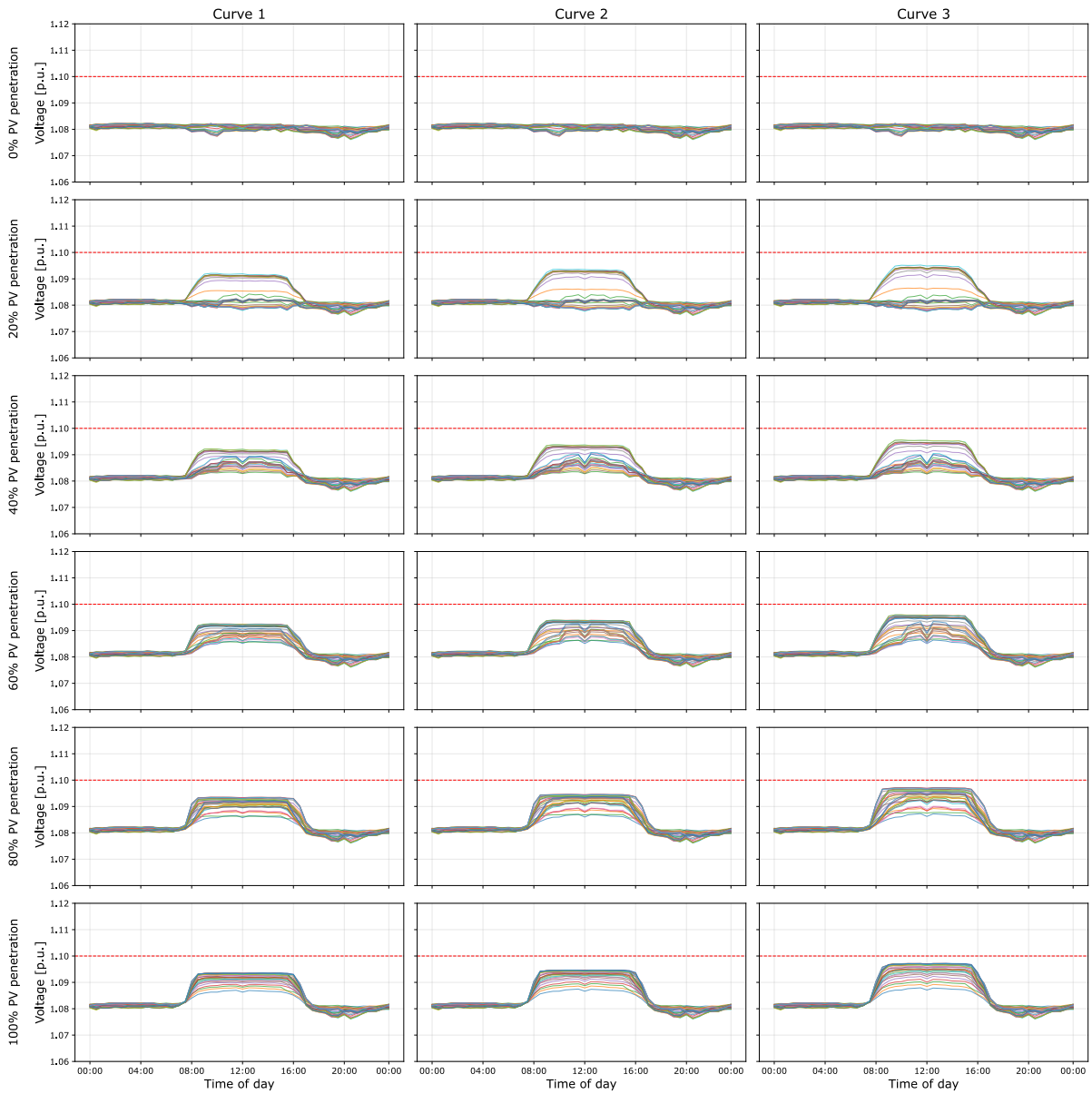

Figure 2: Customer voltage profiles (Autumn day, start_day = 54) for PV penetration levels from 0% to 100% and Volt-Watt Curves 1–3.

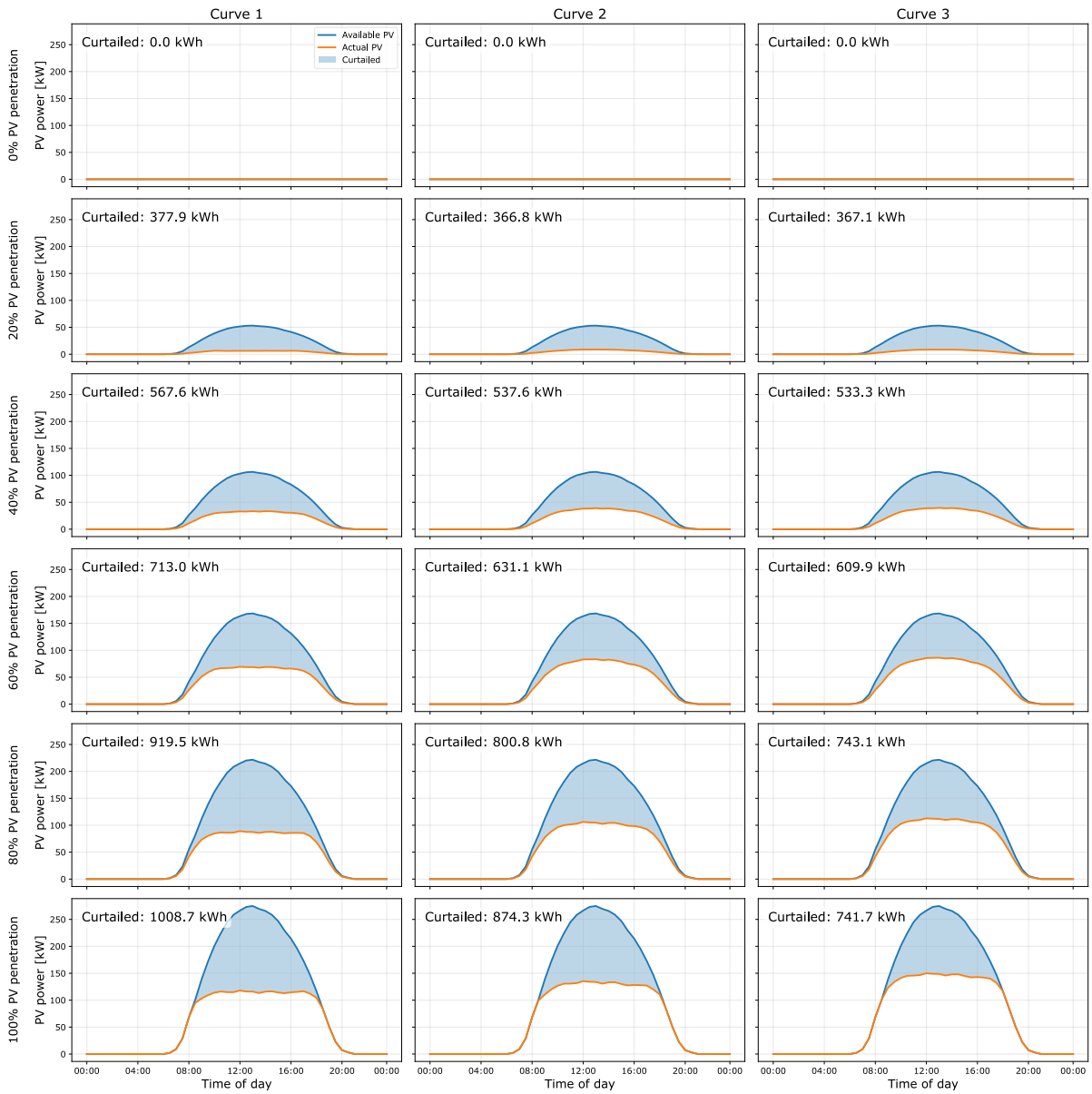

Figure 3: Curtailment results (Summer day, start_day = 28) across PV penetration levels for Volt-Watt Curves 1–3; total curtailed energy shown per subplot.

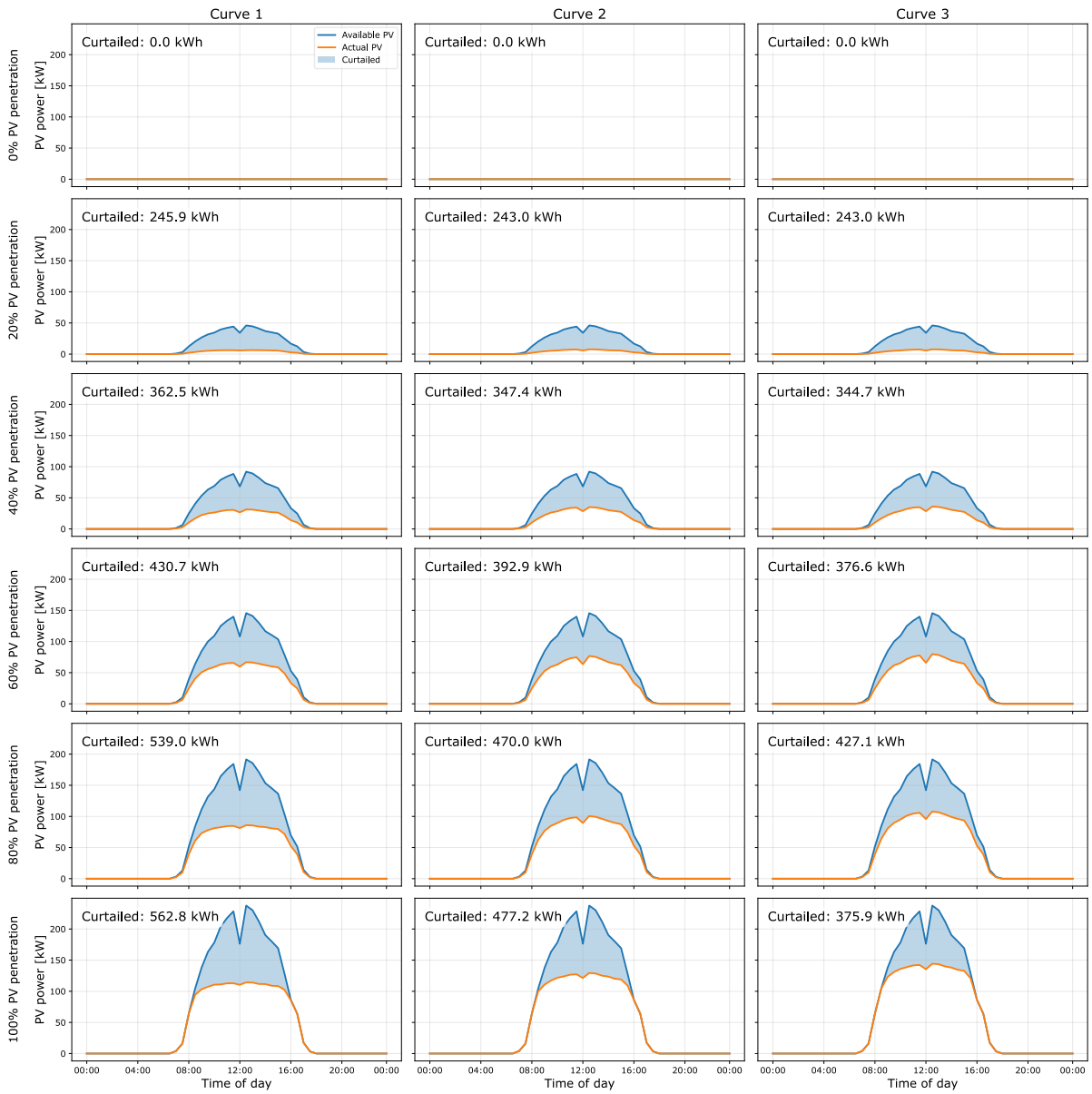

Figure 4: Curtailment results (Autumn day, start_day = 54) across PV penetration levels for Volt-Watt Curves 1–3; total curtailed energy shown per subplot.